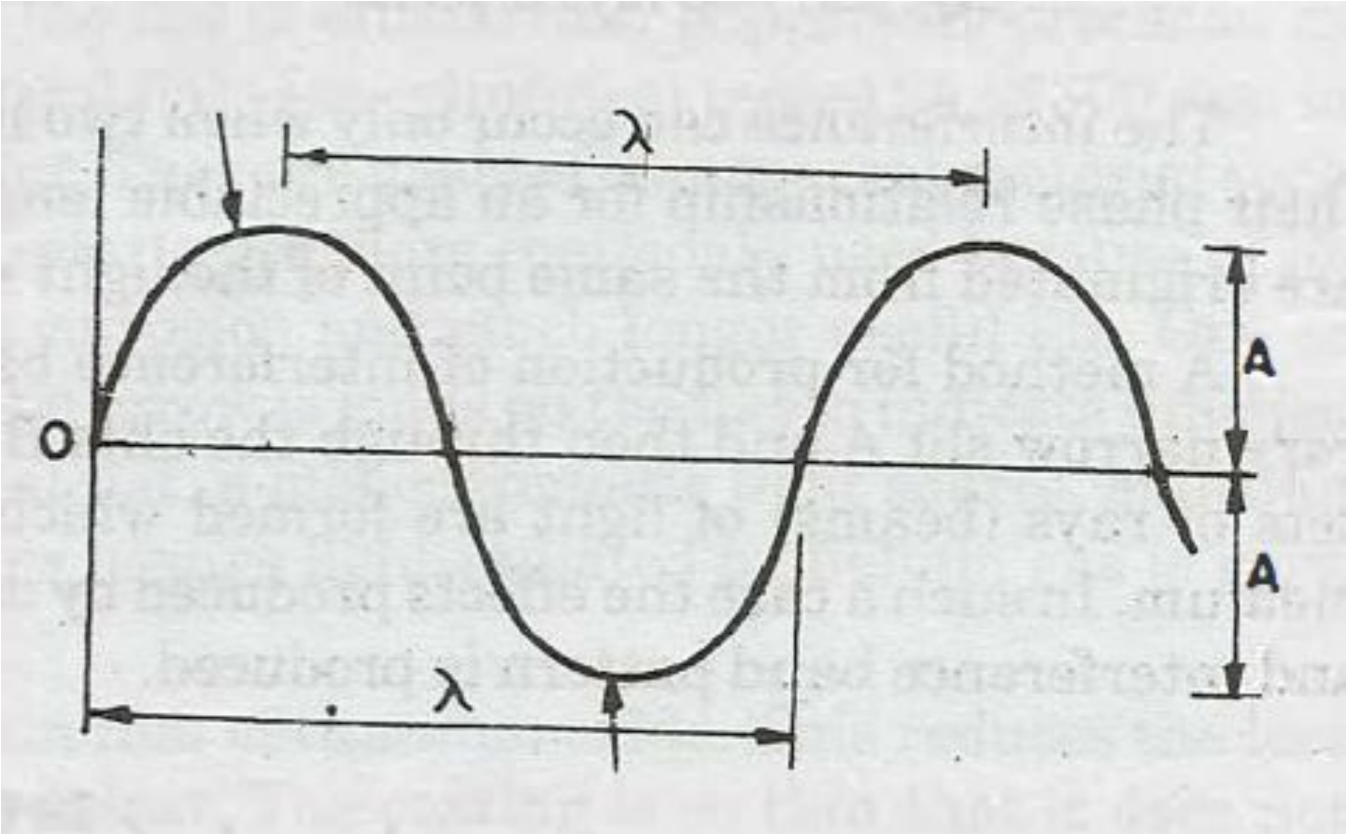
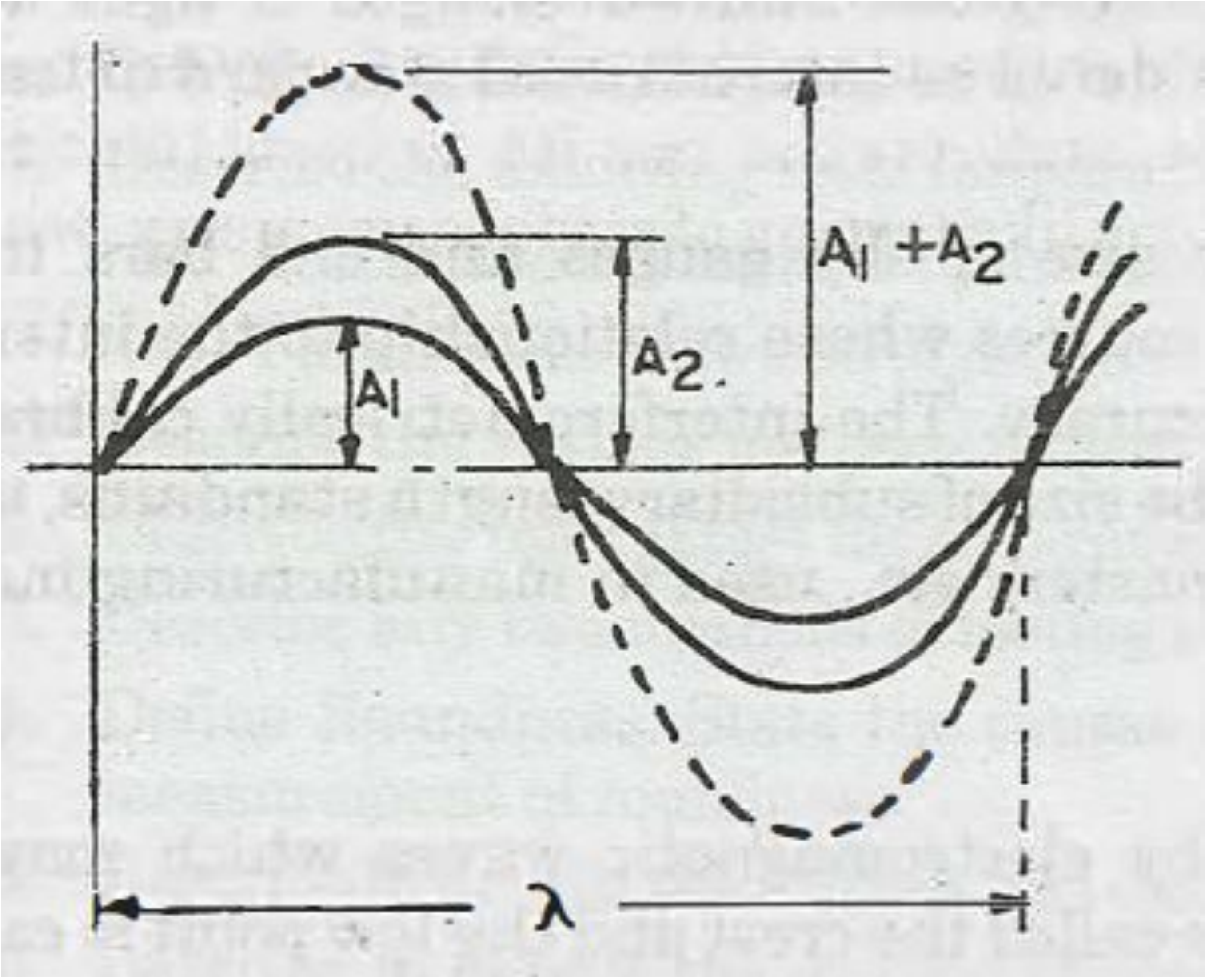
Interferometry

Monochromatic lights



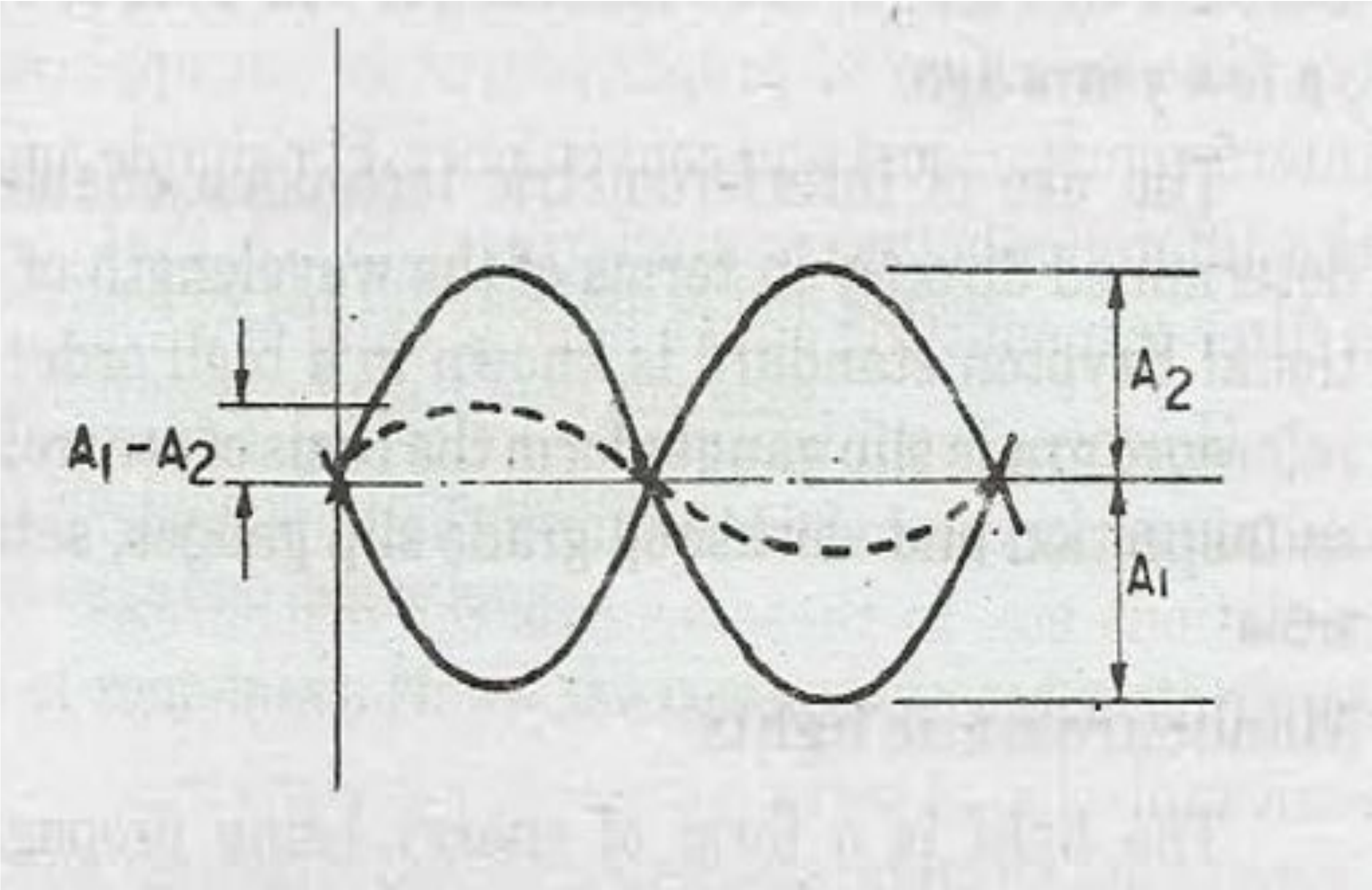
TWO RAYS IN PHASE

PRINCIPLE OF INTERFERENCE

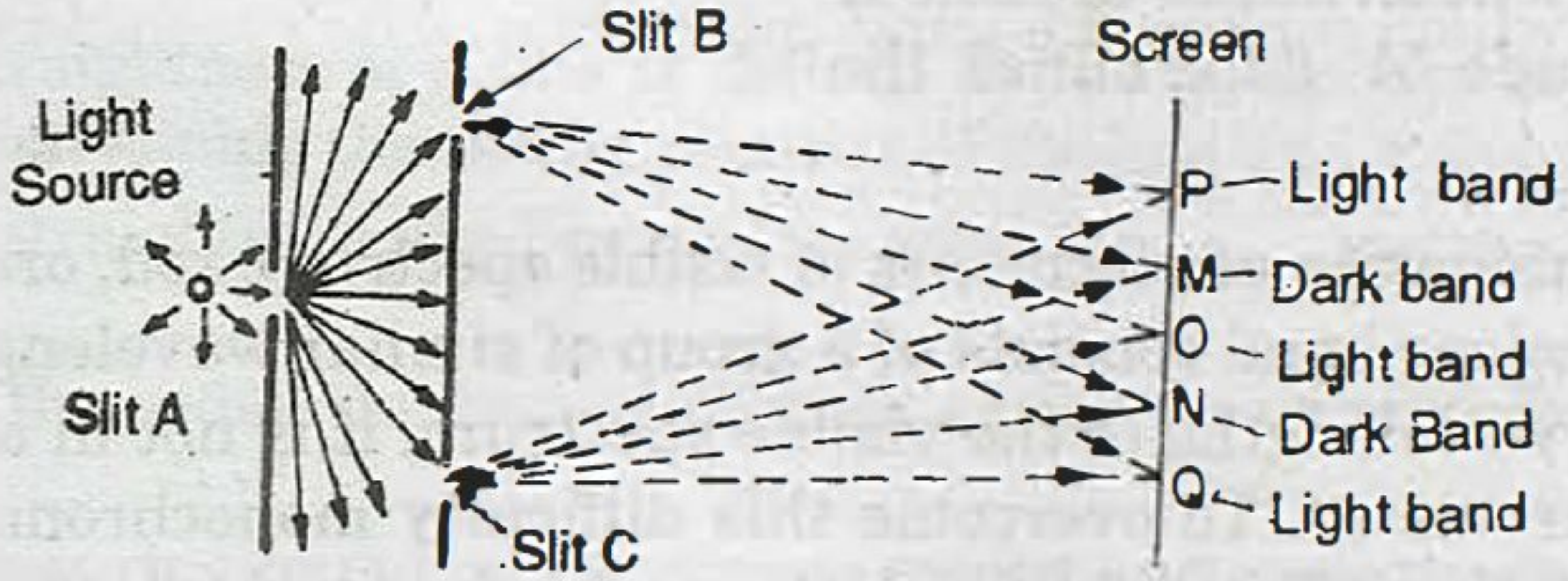


TWO RAYS OUT OF PHASE

PRINCIPLE OF INTERFERENCE



The interference can occur only when two light rays are coherent, that is the two rays maintain their phase relationship for an appreciable length of time. This is possible only when the two rays are originated from the same point of the light source at the same time.

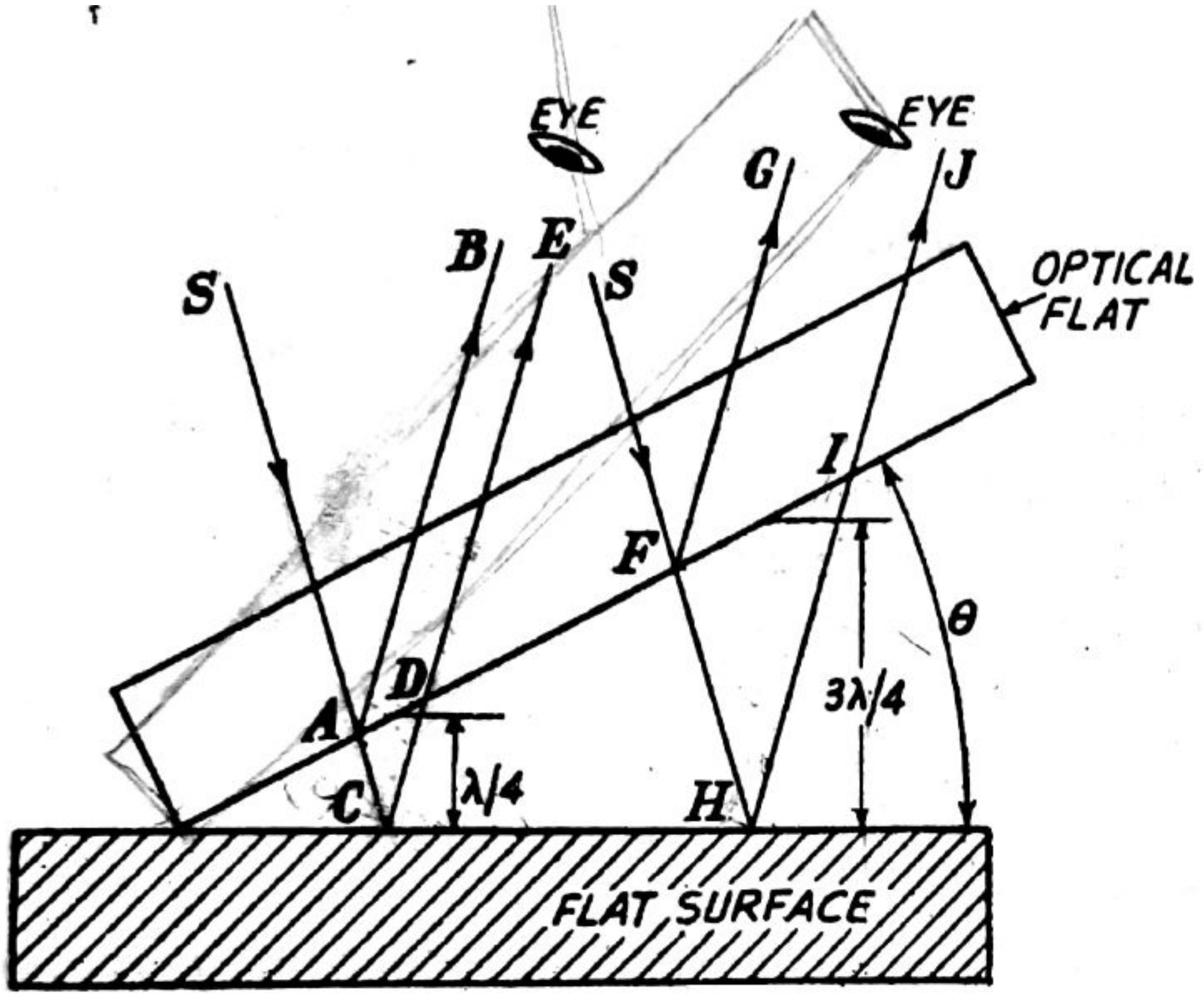

Slit B
Screen
Light Source
P Light band
M Dark band
O Light band
Slit A
N Dark Band
Q Light band
Slit C

If .the paths *BO* and *CO* are exactly equal then the rays on these paths will be in phase. A constructive interference takes place producing the maximum intensity. At some point *M* the ray

path difference will be equal to 1/2 wavelength, *i.e., CM* - *BM*= $\frac{1}{2}\lambda$

Thus at the screen the waves will be 180° out of phase *i.e.,* the resultant intensity will be zero and a destructive interference takes place producing total darkness at Mand at ***N.*** At point ***P,*** the ray path difference will be one wavelength, and the rays again will be in phase and a bright band will be produced. Thus a series of bright and dark bands are produced. These bands are known as interference bands.

Flatness Testing By Interferometry


EYE
EYE
S
B
E
S
G
J
OPTICAL FLAT
I
F
θ
D
A
3λ/4
λ/4
C
H
FLAT SURFACE

At point A, the wave of incident beam from S is partially reflected along AB and is partially transmitted across the air gap along AC.

At C, again the ray is reflected along CD and passes out towards the eye along CDE.

Thus the two reflected components, reflected at A and C are collected and recombined by the eye, having travelled paths whose lengths differ by an amount ACD.

If the path lengths of the two components differ by an odd number of half wavelengths then condition for complete interference is achieved.

If the surface is , perfectly flat, then condition of complete interference is satisfied in a straight line across the surface as the surface at right-angles to the plane of the paper is parallel to the optical flat. Therefore, a straight dark line will be seen passing through point *C*.

Consider another ray incident along path *SFH.*

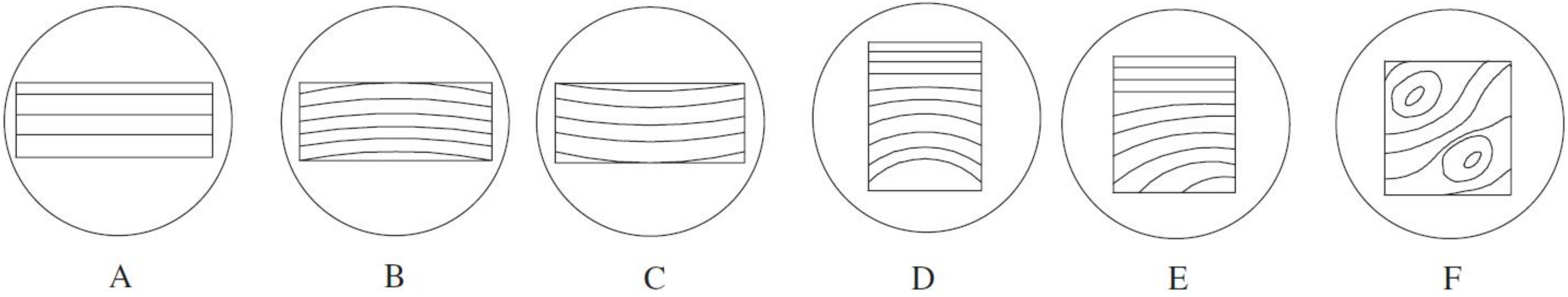
Again this ray is also splitted into two components. It is obvious that the path difference of the two component rays will keep on increasing along the surface due to angle θ. Thus if the path difference *FHI_be* 3λ/2 or the next odd number of half wavelengths, then interference will occur and a similar fringe will be seen.

Next when path difference is 5λ/2, again there will be another dark fringe.

At the intermediate point between the points *C* and *H,* the path difference will be an even number of half wavelengths and the two components will be in phase producing a light band.

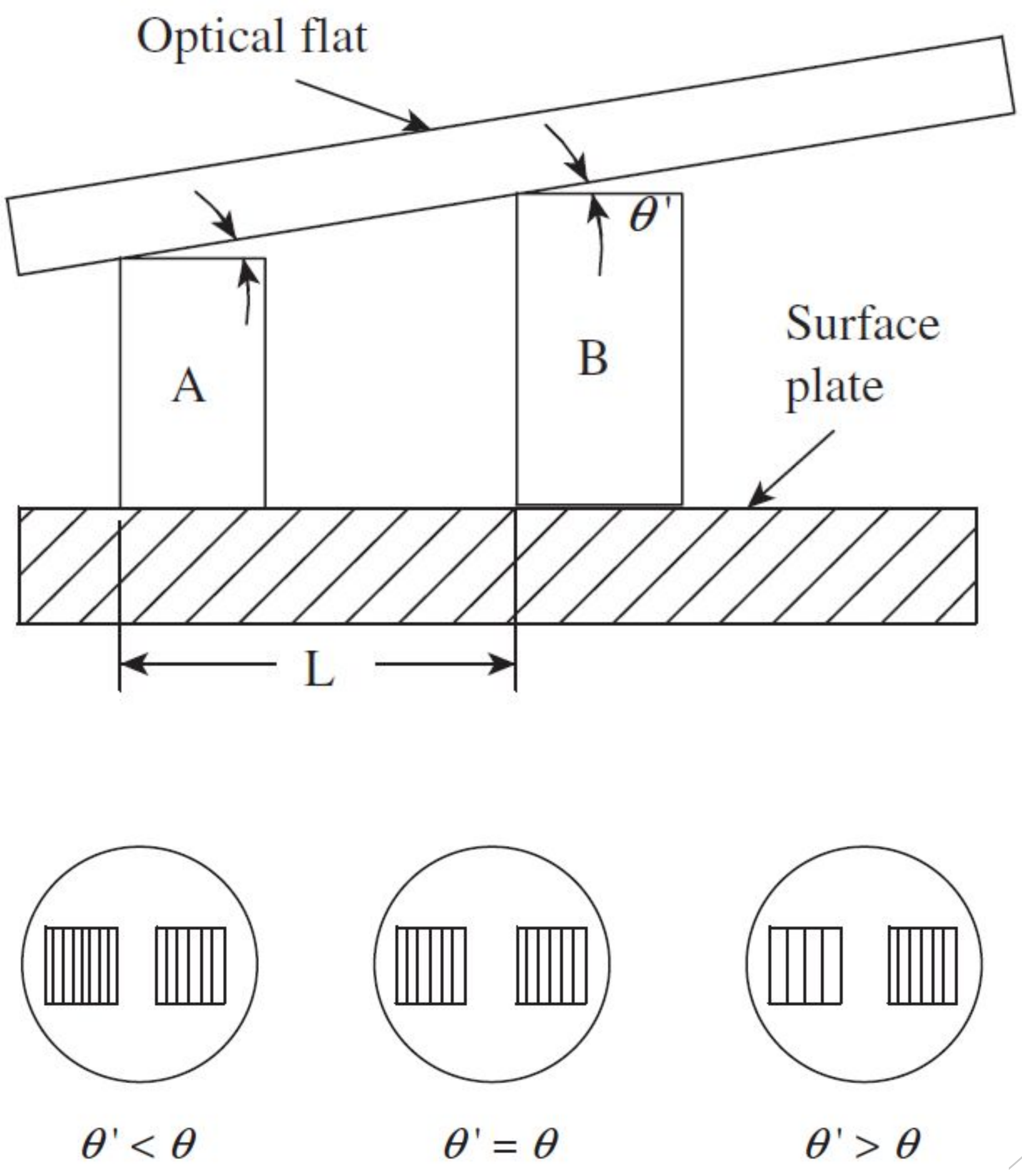
Interference pattern in a perfectly flat surface

Fringe patterns reveal surface conditions



Fringe pattern	Surface condition
A	Block is nearly flat along its length.
B	Fringes curve towards the line of contact, showing that the surface is convex and high in the centre.
C	Surface is concave and low in the centre.
D	Surface is flat at one end but becomes increasingly convex.
E	Surface is progressively lower towards the bottom left-hand corner.
F	There are two points of contact, which are higher compared to other areas of the block.

Comparative Measurement with Optical Flats



Let the number of fringes on the reference block be N over a width of l mm. If the distance between the two slip gauges is L and λ is the wavelength of the monochromatic light source, then the difference in height h is given by the following relation:

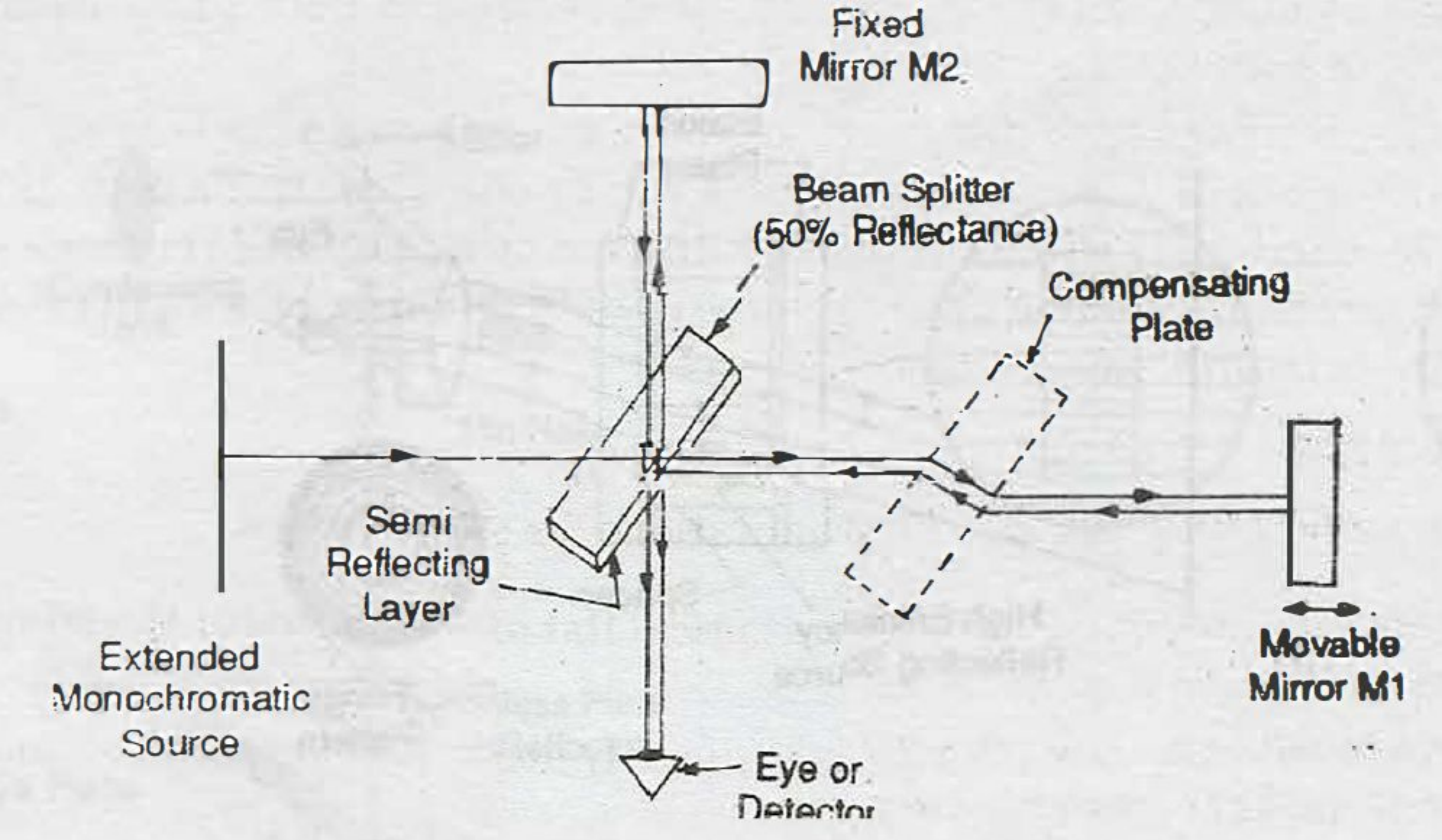
$$h = \frac{\lambda L N}{2l}$$

INTERFEROMETERS

INTERFEROMETERS

Interferometers are optical instruments used for measuring flatness and determining the length of slip gauges.

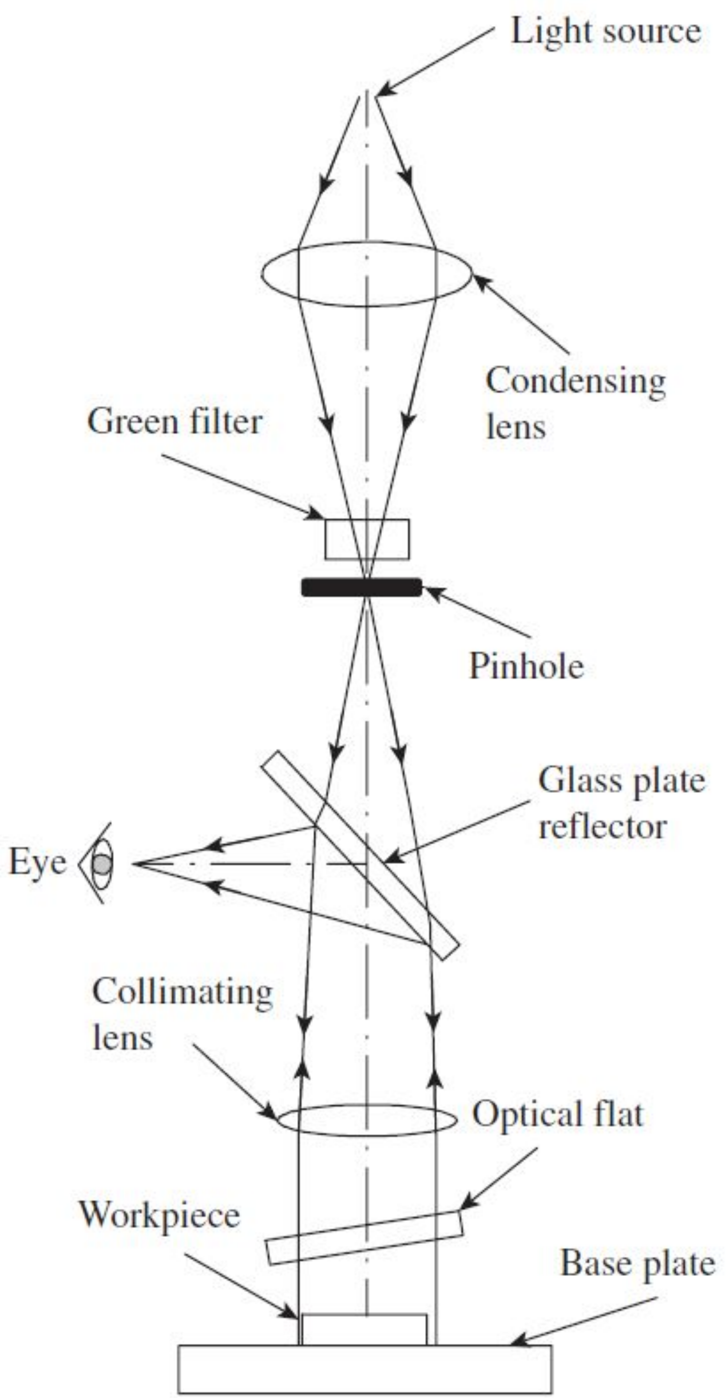
Michelson Interferometer



Michelson Interferometer

The monochromatic light falls on a beam splitter D consisting of semi-reflecting layer. Thus the light ray is divided into two paths. One is transmitted through compensating plate P to the mirror M 1 and the other is reflected through_beam splitter D to mirror M2 . From both these mirrors, the rays are reflected back and these reunite at the semi-reflecting surface from where they are transmitted to the eye and thus the fringes can be observed. Mirror M2 is fixed and mirror M1 is movable i.e., it is attached to the object whose dimension is to be measured.

NPL Flatness Interferometer



NPL Flatness Interferometer

Let us consider a gauge that shows n1 fringes along itslength in the first position and n2 in the second position. The distance between the gauge and the
optical flat in the first position has increased by a distance δ1, over the length of the gauge, and in the second position by a distance δ2. It is clear that the distance between the gauge and the optical flat changes by λ/2, between adjacent fringes.

Therefore, δ1 = n1 * λ/2 and δ2= n2 * λ/2.

The change in angular relationship is (δ2 - δ1), that is, (δ2 - δ1) = (n1 - n2) *λ/2.

The error in parallelism is actually (δ2 - δ1)/2 because of the doubling effect due to the rotation of the base plate. Thus, (δ2 - δ1)/2 = (n1 - n2)/2 *λ/2).

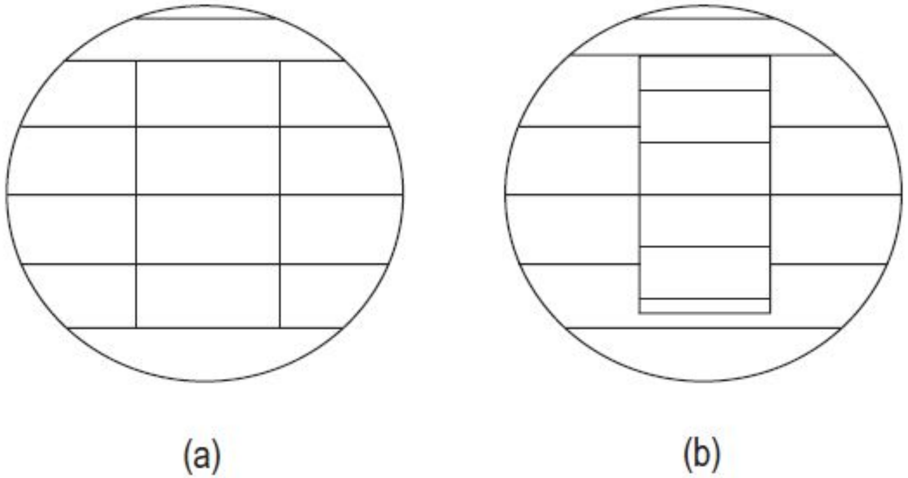


Fig. 7.15 Example of fringe patterns (a) Equal fringes on parallel surfaces (b) Unequal fringes due to flatness error

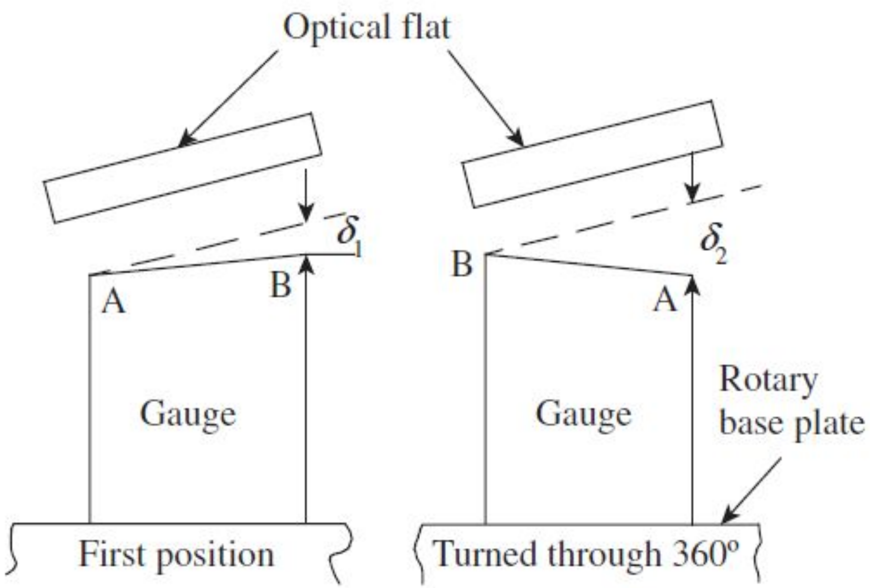
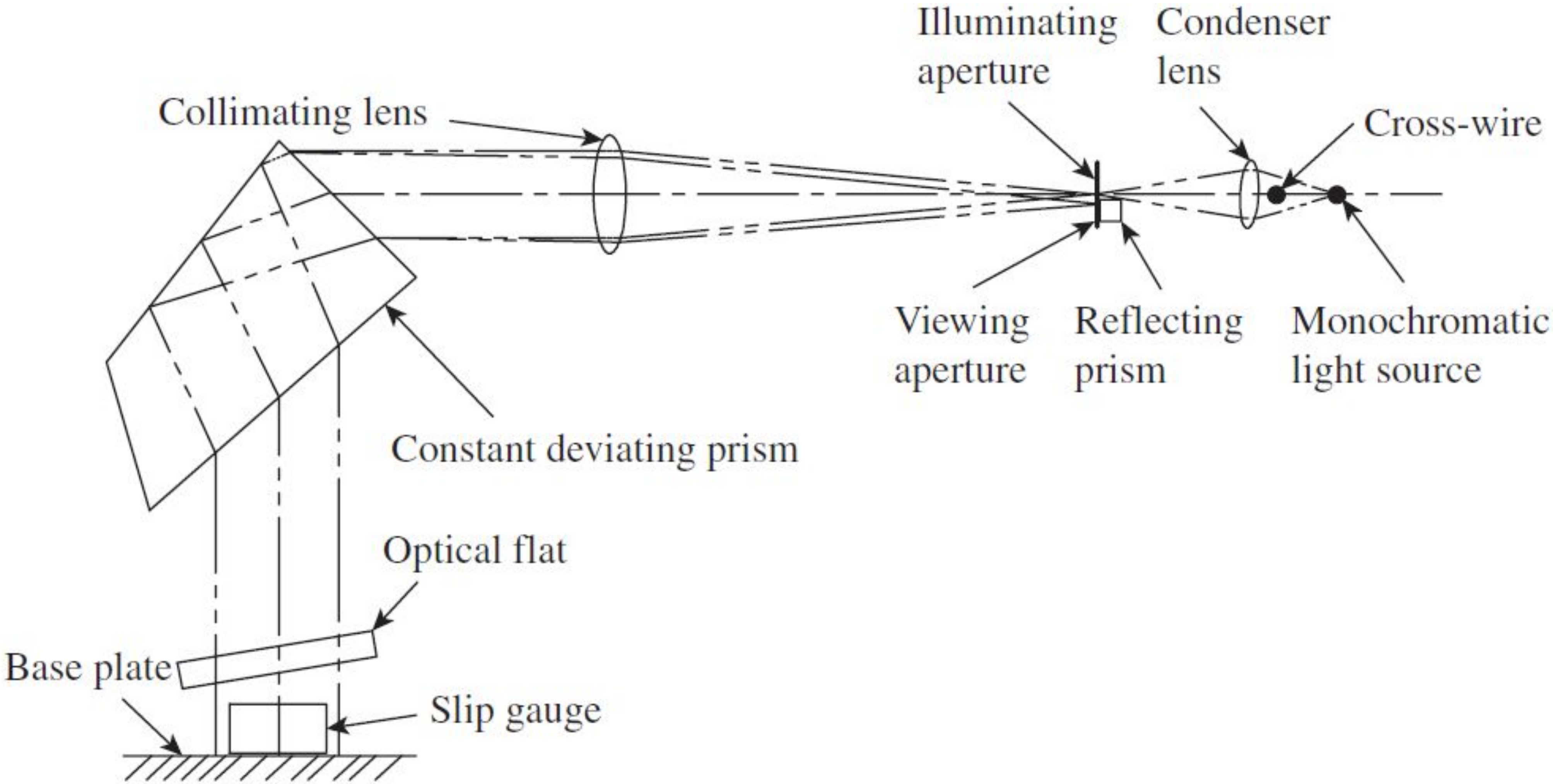


Fig. 7.16 Testing parallelism in gauges

Pitter–NPL Gauge Interferomete r

This interferometer is used for determining actual lengths of slip gauges. Since the measurement calls for a high degree of accuracy and precision, the instrument should be used under highly controlled physical conditions.


Illuminating aperture
Condenser lens
Collimating lens
Cross-wire
Viewing aperture
Reflecting prism
Monochromatic light source
Constant deviating prism
Optical flat
Base plate
Slip gauge